

# Production RAG v1 — Measured Results

An evaluation-first retrieval baseline over 202 official OpenAI and Anthropic documentation pages, with the failures left in.

READ THIS FIRST

**The closed-book control (EXP-NULL) did not run.** No generation credential was available and the provider API host is blocked by this environment's network egress policy. The results file records `status: "blocked"` with the exact error rather than placeholder numbers.

This means the primary question V1 was built to answer — *does retrieval beat what the model already knows?* — remains unanswered. Every retrieval figure below is uncalibrated against the closed-book floor.

## What this measured

V1 is deliberately small: a closed-book control plus four retrieval configurations, with no reranker, no agent loop, no framework orchestration and no live crawler. The point is to find out which retrieval method actually helps before adding anything, and to report the result honestly whether or not it flatters the design.

0.000 → 0.475

Lexical baseline macro recall, before and after diagnosing why it retrieved nothing at all

0.818 vs 0.475

Document-level recall against span-level recall — the gap that caps every configuration tested

## Run identity

|                    |                                                                                                   |
|--------------------|---------------------------------------------------------------------------------------------------|
| Corpus snapshot    | snap_689e336380a054d8039dc35b2c09cd0a                                                             |
| Corpus             | 202 documents / 14,209 chunks — Anthropic 139 docs (12,028 chunks), OpenAI 63 docs (2,181 chunks) |
| Golden set         | 22 cases: 20 retrieval-scored + 2 abstain controls; 22 evidence spans                             |
| Evidence anchoring | (version_id, section_path, char_start, char_end) — never chunk_id                                 |
| Embedding model    | emb_205f51a2d4db0273e121527cb5c6ff83 — offline TF-IDF+SVD, 384 dims, 57.2% explained variance     |
| Retrieval k        | 10, held constant across every experiment                                                         |

## Headline results

`macro recall` averages per-case span recall, so a two-span multi-hop case that finds one span scores 0.5. `fully recalled` is the strict count — every expected span retrieved. `doc recall` credits a span when the correct *document* reached the top 10 even though the wrong chunk came back.

| Experiment                                         | macro recall | fully recalled | spans        | doc recall   |
|----------------------------------------------------|--------------|----------------|--------------|--------------|
| <b>EXP-000 lexical — as shipped</b>                | <b>0.000</b> | <b>0/20</b>    | <b>0/22</b>  | <b>0.000</b> |
| EXP-000 lexical — BM25 (fixed)                     | 0.475        | 9/20           | 10/22        | 0.818        |
| EXP-001 dense (LSA substitute)                     | 0.300        | 5/20           | 7/22         | 0.773        |
| EXP-002 hybrid interleave                          | 0.450        | 8/20           | 10/22        | 0.818        |
| EXP-003 RRF <code>rrf_k=10</code>                  | 0.500        | 9/20           | 11/22        | 0.818        |
| EXP-003 RRF <code>rrf_k=20</code>                  | 0.500        | 9/20           | 11/22        | 0.818        |
| EXP-003 RRF <code>rrf_k=60</code>                  | 0.500        | 9/20           | 11/22        | 0.818        |
| <b>EXP-003 RRF pool 100, <code>rrf_k=60</code></b> | <b>0.600</b> | <b>11/20</b>   | <b>13/22</b> | <b>0.818</b> |

## Paired per-question comparisons

On 20 cases an average is close to meaningless — one question is five points of macro recall. What matters is which specific questions moved.

| Change                                      | rescued | regressed | net       | which questions moved                                              |
|---------------------------------------------|---------|-----------|-----------|--------------------------------------------------------------------|
| EXP-000 lexical → EXP-001 dense             | 1       | 5         | <b>-4</b> | rescued AN-011<br>regressed AN-005, OA-001, OA-002, OA-003, OA-004 |
| EXP-000 lexical → EXP-002 hybrid            | 0       | 1         | <b>-1</b> | regressed OA-003                                                   |
| EXP-001 dense → EXP-002 hybrid              | 4       | 1         | <b>+3</b> | rescued AN-005, OA-001, OA-002, OA-004<br>regressed AN-011         |
| EXP-002 hybrid → EXP-003 RRF k=60           | 2       | 1         | <b>+1</b> | rescued AN-004, OA-003<br>regressed OA-004                         |
| EXP-000 lexical → EXP-003 RRF k=60          | 1       | 1         | <b>0</b>  | rescued AN-004<br>regressed OA-004                                 |
| EXP-000 lexical → EXP-003 RRF k=60 pool=100 | 3       | 1         | <b>+2</b> | rescued AN-004, AN-006, AN-008<br>regressed OA-004                 |

## The uncomfortable findings

### 1. The shipped lexical baseline retrieved nothing at all.

It built its query with `websearch_to_tsquery`, which ANDs every token. A sixteen-word question therefore required all sixteen tokens inside a single chunk, and the `simple` text-search configuration has no stopwords list, so `which`, `does` and `the` were mandatory match terms too. Zero hits on all twenty questions. Replaced with BM25 over the same index: **0.000 → 0.475**.

2. Lexical beats dense, decisively.

BM25 0.475 against 0.300. Switching to dense regressed five of twenty questions and rescued one. This must be read with the embedder caveat in the limitations — it is a statement about *this* embedder, not about dense retrieval.

3. Hybrid interleave is worse than lexical alone.

0.450 against 0.475, net -1 question. Alternating between a strong ranked list and a weak one dilutes the strong one. Interleave was included as a transparent checkpoint before RRF, and it earned its place by failing.

4. RRF's apparent gain over lexical is partial credit, not a win.

RRF posts 0.500 against lexical's 0.475, but per question it is one rescued and one regressed — a wash. The entire macro-recall difference is one extra span on a single two-span multi-hop case.

5. Fusion actively regressed a question lexical got right.

Question OA-004's evidence sits at lexical rank 5 and dense rank 61. RRF averaged the two into rank 13, outside the k=10 cutoff. Fusing a strong list with a weak one costs real answers.

6. rrf\_k is inert until the candidate pool is large enough.

At pool sizes 10–30 all three rrf\_k values give identical results. Pool size dominates. Freezing the pool at the shipped default of 30 would have hidden the best configuration entirely.

| candidate pool | rrf_k=10 | rrf_k=20 | rrf_k=60 |
|----------------|----------|----------|----------|
| 10             | 0.450    | 0.450    | 0.450    |
| 20             | 0.500    | 0.500    | 0.500    |
| 50             | 0.500    | 0.550    | 0.600    |
| 100            | 0.450    | 0.550    | 0.600    |

The ceiling: right document, wrong chunk

Document-level recall is **0.818** for lexical, interleave and RRF alike, against span-level recall of 0.455–0.500. Of the twelve spans the lexical baseline missed, **eight had the correct document in the top ten anyway**.

The identical 0.818 across three different fusion strategies is the tell: fusion is reordering chunks *within an already-correct document set*, not finding documents the individual retrievers missed. That caps what any amount of fusion tuning can buy on this corpus.

ROOT CAUSE

Chunk granularity. One missed answer is a 57-character sentence sitting inside a 3,449-character chunk that dumps every body parameter of an API endpoint — 1.7% of its own chunk. Across the Anthropic half of the corpus, 2,521 of 12,028 chunks exceed 3,000 characters, and the largest reaches 12,341 despite a configured 3,500 budget, because the chunker never splits a single oversized table or code block.

This is diagnosed and **deliberately left unfixed**. Re-chunking is the next experiment, and it must be measured against the same evidence spans rather than folded silently into V1. Because ground truth is anchored above the chunk layer, a re-chunked corpus can be scored against the same 22 unchanged spans.

## Limitations — read before quoting any number above

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1. **EXP-NUL did not run.** Retrieval is uncalibrated against the closed-book floor. This was V1's primary question and it is unanswered.
2. **The dense retriever is a substitute, not an equivalent.** Both the configured sentence-transformer host and the OpenAI embedding endpoint are blocked by network policy, so EXP-001/002/003 use an offline TF-IDF+SVD embedder fitted on the corpus itself. It has no pretrained semantic knowledge and no subword handling. **All dense, hybrid and RRF figures are lower bounds.** Re-running with a real embedding model is the single highest-value next step.
3. **n = 20 is not a holdout.** Nothing here is statistically significant. The paired per-question comparisons are more trustworthy than the averages.
4. **The corpus is provider-skewed.** 139 Anthropic documents to 63 OpenAI, because OpenAI's documentation hosts were unreachable and only its public repositories could be used. Lexical fully recalls 7 of 8 OpenAI cases but only 2 of 12 Anthropic ones; the split tracks document size and structure, not provider quality.
5. **Some parameters were tuned on the evaluation set.** BM25 `k1` and `b` are standard defaults, deliberately not swept. The `rrf_k` and candidate-pool grids were swept on these same 20 cases, so the pool-100 configuration is selected on the evaluation set and is optimistic.
6. **The golden set is single-author.** Every anchor was read against its source document and mechanically verified to contain the quoted claim, but no second annotator reviewed them.

## What V1 has not earned the right to add

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A reranker cannot rescue the worst failure case, because the correct chunk is absent from the top 200 candidates entirely — reranking only reorders what retrieval already found. A larger `k` would inflate the metric without improving retrieval. A similarity threshold addresses a calibration problem the data does not show.

The next experiment justified by evidence is re-chunking, measured against the same unchanged evidence spans. If span recall does not move toward document-level recall after re-chunking, the diagnosis is wrong and reranking becomes the next candidate — but not before.

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Generated from `experiments/summary.json` by `scripts/build_report_pdf.py`. Corpus snapshot `snap_689e336380a054d8039dc35b2c09cd0a`. Raw provider documentation is not redistributed; published results carry evidence anchors, ranks and scores with retrieved text replaced by a content hash.